

Increasing ontological commitment

1. Instrumental

Predictive utility only

Predictive models

2. Object / Role

Components and functions

Causal graphs

3. Subspace

Causal subspaces on $\text{Gr}(k, d)$

Grassmannian geometry

4. Structural

Gauge orbits and holonomy

Fiber bundles

5. Stratified

Strata with varying Whitney stratification

Each layer subsumes the ones below it. Stronger claims require higher commitment.